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Rise of Inspectron: Automated Black-box Auditing of Cross-platform Electron Apps

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Browser-based cross-platform applications have become increasingly popular as they allow software vendors to sidestep two major issues in the app ecosystem. First, web apps can be impacted by the performance deterioration affecting browsers, as the continuous adoption of diverse and complex features has led to bloating. Second, re-developing or porting apps to different operating systems and execution environments is a costly, error-prone process. Instead, frameworks like Electron allow the creation of standalone apps for different platforms using JavaScript code (e.g., reused from an existing web app) and by incorporating a stripped down and configurable browser engine. Despite the aforementioned advantages, these apps face significant security and privacy threats that are either non-applicable to traditional web apps (due to the lack of access to certain system-facing APIs) or ineffective against them (due to countermeasures already baked into browsers). In this paper we present Inspectron, an automated dynamic analysis framework that audits packaged Electron apps for potential security vulnerabilities stemming from developers' deviation from recommended security practices. Our study reveals a multitude of insecure practices and problematic trends in the Electron app ecosystem, highlighting the gap filled by Inspectron as it provides extensive and comprehensive auditing capabilities for developers and researchers.

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